

A project proposal: CBDC Resilience and Real Economic Behavior: Causal Evidence from the DCash Shutdown

Maxwell Woodfield - Queen's University

1. Objective of the research project

The objective of this research project is to identify the causal impact of losing access to a Central Bank Digital Currency (CBDC) on the retail payment ecosystem, specifically, payment behavior, cash holdings, and merchant revenues using the unexpected 2022 DCash service interruption in the Eastern Caribbean Currency Union (ECCU) as a natural experiment.

The DCash outage, caused by a technical issue resulted in all users being unable to make transactions or access their e-Wallets, and provides an valuable setting in which a CBDC was abruptly removed from an economy for 2.5 months (January 14th [Service Interruption](#) to April 4th [Service Resume](#)). Because this applied randomly to users uniformly and across the entire country, it theoretically can constitute a plausibly exogenous shock to individuals, households, and firms who had adopted DCash for transactions (either day-to-day or for short term savings).

This research seeks to answer the central question: "How does the sudden loss of access to a CBDC affect consumers' payment choices, cash demand, and merchant-level economic activity in a partially digitalized monetary environment?"

By providing an empirical measurement of the economic consequences of losing access to a retail CBDC (rCBDC), this study aims to contribute new evidence to the global policy debate surrounding CBDC resilience, adoption, and the importance of digital public money. The results will offer insights into how CBDCs interact with existing payment infrastructures, how quickly users revert to cash or alternative digital platforms under stress, and how payment outages propagate through merchant sales and local economic activity. More broadly, this research can also inform central banks about the operational and behavioral risks associated with CBDC deployment and the conditions under which CBDCs meaningfully shape real economic outcomes.

2. Motivation

The rapid emergence of CBDCs is reshaping global discussions on the future of money, payment-system architecture, and the role of central banks in the digital economy. More than one hundred central banks are actively researching or piloting CBDCs ([CBDC Tracker](#)), yet the empirical evidence base remains thin, especially regarding how CBDCs affect real economic behavior in practice. Despite widespread policy enthusiasm, fundamental economic questions remain open: How do CBDCs change payment choices? Do they meaningfully reduce cash dependency? Do merchants and consumers integrate CBDCs into daily economic activity? And crucially - what happens when CBDC access fails?

The 2022 DCash outage provides a rare opportunity to study these affects through real-world data. The outage, caused randomly by an unexpected technical failure unrelated to economic policy shifts or related changes in behavior presents one of the first exogenous shocks to a live retail CBDCs functionality, resulting in a complete widespread shutdown of all accounts (including the ability to make or receive transactions, or access e-Wallets).

A CBDC hinges on multiple criteria that shape its potential for success (high adoption) and ability to handle transactions. One of these criteria is its resilience, which "refers to a system's ability to avoid loss of service and loss of data if a failure occurs" ([Archetypes for a retail CBDC](#)). A low resilience should theoretically deter individuals from using a rCBDC, especially if this means transactions can randomly be cancelled and wallets inaccessible. However, the effects of such low resilience and abrupt rCBDC usage has yet to be study. The DCash outage demonstrates that failure in rCBDC exist, but the economic consequences of such failures are not well understood. Quantifying these impacts is essential for informing design choices, contingency planning, and risk assessments for future CBDCs

An argument for CBDCs is their potential to reduce dependence on cash and expand digital financial inclusion ([CBDC and Monetary Policy](#)). But if users revert heavily to cash during disruptions, this may suggest that CBDCs may complement rather than replace cash - or that trust and reliability are far more important than technical design assumptions currently acknowledge. Despite this, little evidence exists of how individuals react during a CBDC disruptions or after. This study may provide important insight into consumer behavior towards cash and rCBDC's during or after a disruption to digital currency.

Merchant acceptance is critical for digital currency viability and its usage hinges on whether merchants trust it to act as cash during a transaction. The outage also provides a unique setting to assess whether merchants lost revenue, shifted to alternative payment channels, or faced operational challenges without CBDC access. This is crucial for understanding the business case for CBDC adoption.

While many countries have CBDC pilots, most notable China (e-CNY), none have experienced a well-defined, region-wide, time-limited disruption that permits the causal identification of CBDC impacts in a real retail environment. China's e-CNY has been widely studied, using difference-in-difference models to determine the affects of introducing their CBDC to a new province, it still lacks a comparable discontinuity. Nigeria's eNaira, one of 3 current in use CBDC's ,([CBDC Tracker](#)), has not had "a single outage for the full year" of launch and maintained "24/7 operation" ([Nigeria's eNaira, Pg. 11](#)). The Bahamas' Sand Dollar, and Jamaica's JAM-DEX, have not experienced any long term outage's that allow for a clean identification, and have seen very low adoption due to a lack of government trust and technical issues ([Bahamas' Sand Dollar](#)) ([CBDC Adoption](#)).

This research also provides an opportunity to test economic monetary theory on digital currencies, and whether traditional theory align with empirical data. Theoretically, digital payments reduce time and money costs of payments or withdrawals, from an inventory theoretic approach it is predicted that individuals will substitution away from physical cash and into a lower-cost digital media of exchange ([Transactions Demand for Cash: An Inventory Theoretic Approach](#)). While this motivates the adoption of a CBDC, a real-world outage in a functional CBDC provides evidence of how quickly (or slowly) agents revert to alternative money forms when digital currencies fail, and whether they revert to their original holdings of electronic cash after a system reconstruction or not. Liquidity preferences also sees money as the most liquid asset, if a digital currency offers equal or greater liquidity, with no cost of holdings, this may alter individuals liquidity preferences to non-cash assets, altering portfolio choices and reducing demand for physical currency.

Individuals who do transfer to DCash holdings and away from cash may alter their portfolio preferences before, and after the outage if they believe a digital currency no longer offers them a safe means to hold liquid assets for day-to-day transactions.

3. Literature Review

The literature surrounding digital payments is rapidly expanding as countries begin to introduce CBDC's. Research reflects a growing global interest in how digital public money may reshape payment systems, monetary transmission, and financial inclusion.

Theoretical studies have predicted CBDCs can alter the structure of individuals money holdings, away from bank deposits and towards a CBDC. A CBDC issued in the US "could permanently raise GDP by 3%, due to lower real interest rates, distortionary taxes, and monetary transaction costs" and the "central bank's ability to stabilize the business cycle" ([Barrdear & Kumhof](#)). In addition, a CBDC can deter runs (run risk) and provide the central bank with informational advantages relative to conventional runs into cash ([Brunnermeier & Niepelt](#)). These models predict that a CBDC has the ability to provide liquidity preferences, and deter from cash demands during panics. The DCash outage provides a opportunity to test how robust these theoretical mechanisms are during a digital currency public shutdown, and how preferences to cash (and digital cash) change following the reopening of these e-Wallets.

Recent literature also explores how perceived trust and technological reliability are core determinants of payment-instrument choice ([Szumski \(2020\)](#)). Evidence from U.S. consumers sees cash as being "the most significant substitute to debit cards in retail settings, whereas checks are the most significant in bill-pay settings" ([Koulayev, Rysman \(2015\)](#)). Evidence from card and phone payment outages show large short run increases in cash holdings and away from e-Wallets ([Retail and Financial survey \(2024\)](#)). While CBDC literature highlights design features, such as high reliability and offline capabilities to mitigate fallouts, direct evidence of the impacts on a CBDC shutdown are sparse, with existing work relying on survey experiments or theoretic models.

Empirical research on CBDCs primarily focuses on documenting institutional frameworks rather than behavioral effects. Evidence from China's CBDC (e-CNY) examines adoptions uptake, which is heavily strengthened by regulatory power, economic incentives, and marketing from the government ([Bai, Cong, Luo, Xie \(2025\)](#)). CBDC empirical evidence from China is largely used due to its large-sample sizes, and ability to estimate causal effects by measuring changes to provinces before and after the introduction of pilot programs (primarily through difference-in-difference frameworks). While this evidence provides an understanding of individuals adoption to a CBDC, China has many incumbent e-Wallet and digital retail payment systems already in place, such as AliPay and WeChat Pay, which offer a similar service to a CBDC at the retail level. Low trust in government or technology may shape low adoption to e-CNY but substitution towards cash instead of well-developed retail digital payment system makes measuring individuals cash-demands or liquidity preferences difficult.

This study seeks to provide empirical evidence of a CBDC withdrawal, and quantify how agents payment patterns change, merchants revenue are impacted, the short-term liquidity responses, and changes to cash reliance. The DCash outage provides a unique natural experiment to credible estimate these effects and also test whether the monetary theory behind CBDCs, and theoretical predictions in money demand, liquidity preferences, and payment choice under uncertainty are valid in a real-world setting. This work would also

extend the "payment outage" literature to public digital money, and potentially highlight whether resilience or offline-payments should be a priority in any CBDC project.

4. Methodology and Research Design

We formulate multiple potential methods for identifying the causal impacts on our specified outcome variable, dependent on the exogenous shock to DCash. Common notation among each model framework is specified below:

- Y_{it} is our outcome variable for unit i in time t . This can be merchant revenue, ATM withdrawals, number of active wallets, DCash transaction volume.
- $Post_t$ is our indicator variable, 1 during the outage period, 0 otherwise.
- $Exposure_i$ is a pre-outage measure of how dependent i was to DCash. For instance, this could be the share of wallets in region i , or the pre-outage DCash transaction share for merchant i . This provides us with a indicator of merchant acceptance, and can be encoded as continuous or binary.
- D_{it} measures actual DCash usage by i in period t (ex. transactions).
- X_{it} is our set of covariates, a vector of time-varying controls (tourism arrivals, employment, holidays, etc.)
- α_i is our unit fixed effects
- γ_t is our time fixed effects (day/week)
- ϵ_{it} is our error term

Difference-in-Difference

Our strategy is to exploit cross-sectional heterogeneity in pre-outage DCash exposure. The intuition behind this is that individuals more exposed to DCash (treated) should be more affected by the outage than those less-exposed (control). We then use the outage as our time discontinuity (Post), with exposure defining our treatment intensity:

$$Y_{it} = \alpha_i + \gamma_t + \beta(Exposure_i * Post_t) + X'_{it}\delta + \epsilon_{it}$$

β therefore measures the causal effect of a 1-unit increase in exposure on the change in Y during the outage.

This strategy relies on the assumption of parallel trends (the outcome trends for high and low exposure units would have evolved similarly after conditioning on X_{it} and fixed effects), as well as the assumption that there were no other shocks differentially affecting high-exposure units at the outage date.

2SLS - Outage as Instrument

If DCash usage is endogenous on outcomes, for instance merchant revenue, then we can use an instrumental variable approach with the outage and exposure interaction as an instrument:

First Stage:

$$D_{it} = \alpha_i + \gamma_t + \pi(\text{Exposure}_i * \text{Post}_t) + X'_{it}\phi + \mu_{it}$$

Second Stage:

$$Y_{it} = \alpha_i + \gamma_t + \beta\hat{D}_{it} + X'_{it}\delta + \epsilon_{it}$$

β becomes our LATE, the effect of a marginal reduction in DCash usage (caused by the outage) on Y.

Naturally, this approach relies on the assumption that there is strong enough covariance between D_{it} and $\text{Exposure}_i * \text{Post}_t$, as well as no covariance in error terms and between Y and the outage.

Hypothesis tests for endogeneity ($H_0 : C(\hat{D}, \epsilon) = 0$ vs. $H_A : C(\hat{D}, \epsilon) \neq 0$) can be used to test these assumptions although this relies on homoskedasticity which likely isn't true. A test for residual uncorrelatedness ($H_0 : E(\mu'\epsilon) = 0$) can also be used to check our assumptions.

If our instrument is believed to be weak, we can measure the identification strength with an F-test (although this relies again on homoskedasticity), or consider a LIML approach.

DML or Causal Forest

To avoid strong parametric assumptions about the functional form between outcomes, treatments and covariates we can use a double machine learning or generalized causal forest to estimate a ATE or CATE. This design may be superior to 2SLS or DiD if we have a high dimension of control vectors that follow a highly nonlinear functional form.

Our treatment variable is again, $T_{it} = \text{Exposure}_i * \text{Outage}_t$ where outage is 1 during DCash outage dates and 0 otherwise, while Exposure is an individual's pre-outage intensity measure constructed from a merchant's revenue share from DCash or transaction frequency.

The model becomes:

$$Y_{it} = \theta T_{it} + g(X_{it}) + \epsilon_{it}$$

Where θ is our causal ATE of outage exposure, and $g()$ is an unknown nonlinear function.

Our treatment model is:

$$T_{it} = m(X_{it}) + \mu_{it}$$

We then use a sample splitting and cross-validation (either K-folds or LOOCV depending on computational cost) and estimate:

$$\hat{g}^{(-k)}(X_{it}) = E[Y_{it}|X_{it}]$$

$$\hat{m}^{(-k)}(X_{it}) = E[T_{it}|X_{it}]$$

Using some machine learning algorithm (random forest, neural network, gradient boosting, etc.) and compute the residuals on our held-out fold to get:

$$\tilde{Y}_{it} = Y_{it} - \hat{g}^{(-k)}(X_{it})$$

$$\tilde{T}_{it} = T_{it} - \hat{m}^{(-k)}(X_{it})$$

Our final stage regression becomes $\tilde{Y}_{it} = \theta \tilde{T}_{it} + u_{it}$ where $\hat{\theta}$ is our ATE of outage exposure.

To estimate heterogeneity treatment effects, we can estimate a CATEs through a causal random forest framework where we grow a random forest using splits designed to maximize the variance in treatment effects. This approach can be useful for estimating which merchants suffered most from the outage (say, in terms of revenue changes), or heterogeneity by tourism and employment intensity.

For example, this can be applied to our merchant revenue as:

$$T_{it} = DCashShare_i^{pre} * Outage_t$$

$$Y_{it} = \log(Revenue_{it})$$

With the DML method providing ATE (average % revenue loss per unit exposure) and the CRF identifying which merchants lost the most.

These methods all rely heavily on the data available, with optimal estimation strategies from our list above depending on the assumptions we can make about our model (of particular importance for DiD and 2SLS), the number of strong covariates we have data on (necessary to map high dimensional nonlinear interactions for DML or CRF with ML) and what outcomes variables we have access to (ex. merchant revenues vs. atm withdrawals vs. DCash transactions, etc.). Further methodology and design should be reconsidered after the data collection process is complete.

5. Implementation Plan

Research Steps:

1. Framework and Hypothesis'

After providing a introduction to our topic, including the history of the Caribbean's DCash CBDC program from its inception to current development, as well as our literature review related to our project, we shall formalizing a clear hypothesis. Our hypothesis, based on monetary theory from liquidity preferences, transaction costs, payment technology adoption frameworks, and trust in technology will theorize the expected findings from our research. Based on payment substitution between CBDC and cash or digital alternatives, liquidity preferences and cash holdings, merchant revenues and transaction volumes our hypothesis' will consist of a testable set of positive questions linking CBDC withdrawal to measurable behavioral outcomes.

Working Hypothesis':

During outage:

H_1 : During the DCash outage, regions and merchants with higher pre-outage DCash reliance experience larger increases in cash withdrawals and greater usage of alternative private digital payment rails.

H_2 : Regions with higher pre-outage DCash penetration experience a statistically significant increase in net cash withdrawals during the outage period.

H_3 : Merchants that relied more heavily on DCash prior to the outage experience a causal decline in total revenues and transaction counts during the outage period.

Post outage:

H_4 : After DCash services are restored, usage does not return to pre-outage levels, particularly among users and merchants with higher pre-outage exposure.

H_5 : The share of inactive DCash wallets increases in the post-outage period, reflecting user attrition and loss of confidence in the system.

2. Data Collection and Cleaning

Potential data sources include: ECCB transaction data over the entirety of DCash's history (including e-Wallet deposits, transactions between merchants and customers, number of e-Wallets opened, number of e-Wallets unused, changes in acceptance of DCash by merchants, and total transaction volumes), cash withdrawals (including ATM withdrawals, cash deposits), card or alternative mobile-money usages, merchant point of sale data from partner providers, potential confounding regional factors to control for (tourism, employment), and a complete timeline of the ECCB outage (from date of outage, to date of reopening).

After collection this data, we shall then construct a tidy panel dataset (likely by daily frequency) and labelling for the pre-outage, outage, and post-outage periods.

3. Descriptive Analysis

This will include summary statistics and plotted time-series for each of our variables with indicators for there 'treatment group' (pre-outage, outage, post-outage). Identifying visible substitution patterns, timings, and accuracy checks for the validating of causal inference.

4. Empirical Identification Strategy

Methods for identification will depend on our data's structure, causal validity strength, and assumed underlying true form. Suggested methods for identification include those discussed in part 4 above including robustness checks and interpretations of potential results.

Challenges:

Our primary challenge lies in data collection. To estimate the effects of the CBDC outage on multiple outcome variables listed above, we will need transaction data from DCash users which the central bank may not grant us access to. Restrictions to data on merchant revenues, ATM withdrawals and DCash uptake hinder our ability to formulate a proper research design, as also noted in our methodology section. Our estimate strategy also relies heavily on strong DCash penetration into the Caribbean market, with low uptake across all regions and merchants, the effects of the outage may have little to no observable difference in outcome variables making causal inference impossible. If pre-outage adoption data fluctuates negligibly or is noisy (measurement error), obtaining significant estimation coefficients will also be difficult.

Confounding economic events, such as random shocks to the Caribbean economy around the 2 month outage (especially those that are difficult to control for), can make estimation strategies difficult, and will need to be checked thoroughly to ensure the validity of our estimates.

While our estimates (if feasible) can provide a causal estimate of the effects of a CBDC withdrawal, these effects are still specific to a Caribbean context and as such can not be generalized globally but, provide empirical evidence of the risks of a CBDC outage on individuals.

Conjecture of Research Findings

Based on the existing evidence in our literature review and motivation, we anticipate several plausible findings:

1. Sharp but temporary shift towards cash

- ATM withdrawals likely spike shortly after DCash becomes unavailable, as people revert back to cash as their e-Wallets become unavailable
- Households will begin to hold more cash after DCash outage as a precaution due to a reliability concern in the government
- After the reopening of DCash, individuals who had opened accounts substitute away from cash to DCash
- Note, we expect this from individuals who used DCash before the treatment, with non users (control) keeping preferences the same

2. Long term shift towards cash

- Similar to the case above, individuals shift to cash during the outage
- However, after the outage DCash usage stays low (perhaps the same as during outage), and individuals stay holding cash, and using other substitutes due to low reliability in CBDC and government
- Note, we expect this from individuals who used DCash before the treatment, with non users (control) keeping preferences the same

3. Substitute toward existing digital rails

- Card payments and mobile-money transfers increase as temporary substitutes
- The extent of substitution depends on merchant acceptance and consumer familiarity
- After reopening, individuals will not switch back to DCash, moving funds from their e-Wallets to the substitute (the opposite is possible but highly unlikely as commercial banks generally offer similar services to a rCBDC, and with the reduced reliance in a CBDC, individuals would need an incentive)

4. Merchant revenue decline among high-DCash users

- merchants that rely heavily on DCash will experience lower sales by disrupted transactions
- the downward spike in revenue should shift back to pre-outage levels during the outage, if they quickly begin accepted other payment methods
- all merchants part of a government program that uses DCash transactions should seen downward revenue spikes in the short run

5. Heterogeneous treatment effects

- regions with strong mobile connectivity substitute toward digital alternatives faster (assuming pre-treatment rates are similar among regions)
- areas with higher trust in government resort back to DCash faster than those without

- small merchants using DCash for low-cost transactions lose more revenue

6. Persistent effects on trust and usage

- After the reopening of DCash, usage will likely remain lower than pre-outage levels
- Trust in CBDC falls and long-term adoption becomes slow